

03_ml_01_2025s1: ML1 (RF and GridSearchCV)

This homework is a classification task to identify whether a mushroom is edible or poisonous. This dataset includes descriptions of hypothetical samples corresponding to 23 species of gilled mushrooms in the *Agaricus* and *Lepiota* Family Mushroom drawn from The Audubon Society Field Guide to North American Mushrooms (1981).

Each species is identified as definitely edible, definitely poisonous, or of unknown edibility and not recommended. This latter class was combined with the poisonous one. The Guide clearly states that there is no simple rule for determining the credibility of a mushroom; no rule like "leaflets three, let it be" for Poisonous Oak and Ivy.

Problems

Please complete the class **MushroomClassifier** using the provided code template. The goal is to differentiate between edible and non-edible mushrooms. The details are as follows:

1. Load 'mushroom2020_dataset.csv' data from the "Attachment" (note: this data set has been preliminarily prepared.).
2. Drop rows where the target (label) variable is missing.
3. Drop the following variables: 'id', 'gill-attachment', 'gill-spacing', 'gill-size', 'gill-color-rate', 'stalk-root', 'stalk-surface-above-ring', 'stalk-surface-below-ring', 'stalk-color-above-ring-rate', 'stalk-color-below-ring-rate', 'veil-color-rate', 'veil-type'
4. Examine the number of rows, the number of digits, and whether any are missing.
5. Fill missing values by adding the mean for numeric variables and the mode for nominal variables.
6. Convert the label variable e (edible) to 1 and p (poisonous) to 0 and check the quantity.
class0: class1
7. Convert the nominal variable to numeric using a dummy code with drop_first = True.
8. Split train/test with 20% test, stratify, and seed = 2020.
9. Create a Random Forest with GridSearch on training data using 5 CV, n_jobs=-1, and scoring='f1_weighted' with the following parameters:
 - 'criterion':['gini','entropy'] ,
 - 'max_depth': [2,3] ,
 - 'min_samples_leaf':[2,5] ,
 - 'N_estimators':[100] ,
 - 'random_state': 2020

10. Predict the testing data set with classification_report.

And return the output based on the question number:

- For Q1, following step 1, please **return an integer number of "na" are there in "gill-size" variables** before doing the data prep.
- For Q2, following step 2-4, please **return a tuple of rows of data and variables**.
- For Q3, following steps 5-6, please **return a tuple of quantity of class0 and class1 in the format as follows (class0_quan, class1_quan)**.
- For Q4, following steps 7-8, please **return a tuple of shape of training (X_train) and testing set (X_test)**.
- For Q5, following step 9, please **return a tuple of best params in the format as follows (criterion, max_depth, min_samples_leaf, n_estimators, random_state)**.
- For Q6, following step 10, please inspect the classification report and **return only F1-score of class 0 and 1 with 2 digits in the format as follows (F1_class0, F1_class1)**.

Submission: **** When submitting to the grader, submit ONLY libraries, class MushroomClassifier with your modified functions.****

Expected Results

Input	Output
print(hw.Q1())	121
print(hw.Q2())	(5764, 12)
print(hw.Q3())	(3660, 2104)
print(hw.Q4())	((4611, 42), (1153, 42))
print(hw.Q5())	('gini', 3, 2, 100, 2020)
print(hw.Q6())	(0.98, 0.97)

Template codes

```

class MushroomClassifier:
    def __init__(self, data_path): # DO NOT modify this line
        self.data_path = data_path
        self.df = pd.read_csv(data_path)

    def Q1(self): # DO NOT modify this line
        """
            1. (From step 1) Before doing the data prep., how many "na" are there in
            "gill-size" variables?
        """
        # remove pass and replace with you code
        pass

    def Q2(self): # DO NOT modify this line
        """
            2. (From step 2-4) How many rows of data, how many variables?
        """
        # remove pass and replace with you code
        pass

    def Q3(self): # DO NOT modify this line
        """
            3. (From step 5-6) Answer the quantity of class0 and class1
            - Note: You need to reproduce the process (code) from Q2 to
            obtain the correct result.
        """
        # remove pass and replace with you code
        pass

    def Q4(self): # DO NOT modify this line
        """
            4. (From step 7-8) How much is each training and testing sets
            - Note: You need to reproduce the process (code) from Q2, Q3 to
            obtain the correct result.
        """
        # remove pass and replace with you code
        pass

    def Q5(self): # DO NOT modify this line
        """
            5. (From step 9) Best params after doing random forest grid search.
            Create a Random Forest with GridSearch on training data with 5 CV,
            n_jobs=-1, scoring='f1_weighted' and the parameters as follow:
            - 'criterion':['gini','entropy']
            - 'max_depth': [2,3]
            - 'min_samples_leaf':[2,5]
            - 'N_estimators':[100]
            - 'random_state': 2020
            - Note: You need to reproduce the process (code) from Q2, Q3, Q4 to obtain
            the correct result.
        """
        # remove pass and replace with you code
        pass

    def Q6(self): # DO NOT modify this line

```

"""

6. (From step 10) What is the value of macro f1 (2 digits)? Predict the testing data set with `classification_report`, using scientific rounding (less than 0.5 dropped, more than 0.5 then increased)
- Note: You need to reproduce the process (code) from Q2, Q3, Q4, Q5 to obtain the correct result.

"""

remove pass and replace with you code
pass